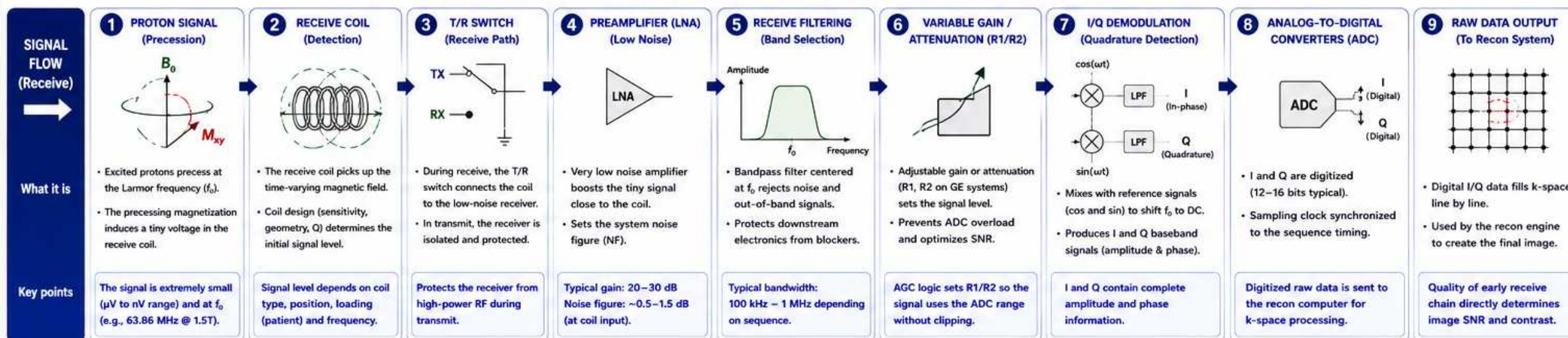




The path of the MR signal during receive: Proton signal → Coil voltage → Low-noise amplification → Filtering & conditioning → Digitization → Raw data

GOAL: Capture the faint MR signal with maximum fidelity and convert it into clean digital data for image reconstruction.



ANALOG SIGNAL LEVELS (TYPICAL)

Point in Chain	Typical Signal Level
At coil (preamp input)	0.1 μV – 10 μV
After preamplifier output	1 mV – 100 mV
After filtering	1 mV – 100 mV
After R1/R2 (receiver in)	10 mV – 1 V
At ADC input (full scale)	$\sim 1 \text{ Vpp}$ differential

These are approximate values and vary by system, coil, sequence, patient, and bandwidth.

WHY LOW NOISE MATTERS

The MR signal is extremely small.

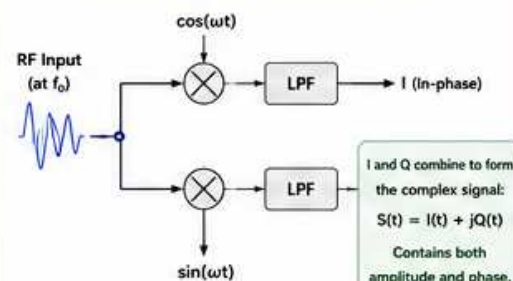
Noise added early cannot be removed later.

The preamplifier **sets the system noise figure**.



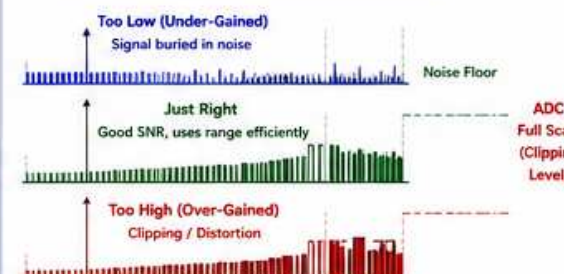
I/Q DEMODULATION EXPLAINED

Shifts the RF signal at f_c to baseband (0 Hz)



RECEIVER GAIN (R1/R2) – AGC WINDOW

Set to maximize dynamic range without clipping.



COMMON NOISE SOURCES

- Patient thermal noise (dominant)
- Preamplifier noise
- Electronic noise (resistors, mixers, ADCs)
- Environmental RF interference
- Gradient switching noise (reduced by proper filtering & blanking)

Good filtering, shielding, and grounding keep noise low and SNR high.

WHAT ENGINEERS MONITOR (SERVICE / ENGINEERING PAGES)

- Raw noise (pre-scan noise measurement)
- Noise correlation (I/Q balance)
- Receiver gain settings (R1, R2)
- ADC utilization / headroom
- Coils element status (array coils)
- Signal amplitude during prescan (TG/RG calibration)
- Overload / clipping indicators

Example (GE Engineering Page)	
Raw Noise (RMS)	0.75 μ
R1	14 dB
R2	22 dB
ADC Utilization	68 %
I/Q Balance (ϵ)	1.03 %
Coil Elements Active	16 / 16
Overload Events	0
Status: OK	

GOOD SIGNAL vs POOR SIGNAL (IMPACT ON IMAGE)

GOOD RECEIVE CHAIN
(High SNR)



- High SNR
- Good contrast
- Sharp detail
- Fewer artifacts

POOR RECEIVE CHAIN
(Low SNR)



- Low SNR
- Poor contrast
- Noise / grain
- Reduced detail

TROUBLESHOOTING GUIDE (RECEIVE CHAIN)

Symptom / Finding	Possible Cause	What to Check
Low SNR (all sequences)	Poor coil, cable, or preamp	Coil elements, cables, preamp noise
Low SNR (only some channels)	Failed coil element	Array coil element status
Noise high	Preamplifier issue or bad connection	Noise measurement, connectors
Clipping / Overload	R1/R2 too low (under-attenuated)	Increase R1/R2 / check AGC
Image shading	Bad coil element or preamp	Element-by-element check
Zipper / RF interference	Cable shield / grounding issue	Cables, routing, shielding

QUICK CHECK FLOW

- 1 Check coil & cables
- 2 Check preamp noise
- 3 Check raw noise level
- 4 Check R1/R2 settings
- 5 Check for clipping / overload
- 6 Verify element status



- The receive chain must be quiet, linear, and stable.
- Small problems in the front end cause big image problems.
- Always use proper handling and ESD precautions for coils and preamps.

ABBREVIATIONS

RF = Radio Frequency
LNA = Low Noise Amplifier
AGC = Automatic Gain Control

R1/R2 = Receiver attenuation settings (GE systems)
ADC = Analog-to-Digital Converter
SNR = Signal-to-Noise Ratio

I = In-phase (real component)
Q = Quadrature (imaginary component)
NF = Noise Figure



THE BOTTOM LINE

A clean, well-optimized receive chain captures the weakest signals with maximum fidelity → higher SNR → better images.

Good Receive Chain Performance = High SNR, Good Contrast, Fewer Artifacts, and Reliable Quantitative Results.